

Problem 1. Consider the matrix

$$\begin{pmatrix} 9 & 0 \\ 0 & 15 \end{pmatrix}.$$

Write a sequence of $\mathbb{Z}$ -linear row and column operations to get the matrix

$$\begin{pmatrix} 3 & 0 \\ 0 & 45 \end{pmatrix}.$$

Then explain a general method to transform a matrix

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

to

$$\begin{pmatrix} \gcd(a, b) & 0 \\ 0 & \text{lcm}(a, b) \end{pmatrix}, \text{ whenever } a, b \text{ are elements of a PID } R.$$

Proof. First, observe that $\gcd(a, b) = \frac{ab}{\text{lcm}(a, b)}$. Let $d = \gcd(a, b)$. Then $a = \alpha d$ and $b = \beta d$ for some $\alpha, \beta \in R$. Also, $\text{lcm}(a, b) = \alpha\beta d$.

Now suppose $\delta \mid \alpha, \beta$. Then $a = \alpha d$ and $b = \beta d$, so $\delta d \mid a, b$. Hence $\delta d \mid \gcd(a, b) = d$, so δ is a unit. Thus $\gcd(\alpha, \beta) = 1$. Since R is a PID, there exist $x, y \in R$ such that $x\alpha + y\beta = 1$. Multiplying by d , we get $xa + yb = d$.

We use this relation to transform the matrix. Starting with $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$, add column 2 to column 1 to get $\begin{pmatrix} a & 0 \\ b & b \end{pmatrix}$. Then use the Euclidean process on the entries in column 1 until the first column becomes $\begin{pmatrix} d \\ 0 \end{pmatrix}$. Then we get a matrix of the form $\begin{pmatrix} d & u \\ 0 & v \end{pmatrix}$. Then subtract multiples of column 1 from column 2 to clear u , obtaining $\begin{pmatrix} d & 0 \\ 0 & v \end{pmatrix}$. We can do this since u was obtained from multiples of a and b both of whom share d as a divisor.

Every row and column operation preserves the ideal generated by all entries, so $d = \gcd(a, b)$ remains the first diagonal entry. Also row and column operations change determinant only by multiplication by a unit. Therefore, ab is associate to dv and v is associate to $\frac{ab}{d} = \text{lcm}(a, b)$. So the final matrix is equivalent to $\begin{pmatrix} \gcd(a, b) & 0 \\ 0 & \text{lcm}(a, b) \end{pmatrix}$.

For the specific matrix,

$$\begin{pmatrix} 9 & 0 \\ 0 & 15 \end{pmatrix} \xrightarrow{C_2 = C_1 + C_2} \begin{pmatrix} 9 & 0 \\ 15 & 15 \end{pmatrix} \xrightarrow{R_2 = R_2 - R_1} \begin{pmatrix} 9 & 0 \\ 6 & 15 \end{pmatrix} \xrightarrow{R_1 = R_1 - R_2} \begin{pmatrix} 3 & -15 \\ 6 & 15 \end{pmatrix} \xrightarrow{R_2 = R_2 - R_1} \begin{pmatrix} 3 & -15 \\ 3 & 30 \end{pmatrix} \\ \xrightarrow{R_1 = R_1 - R_2} \begin{pmatrix} 0 & -45 \\ 3 & 30 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 3 & 30 \\ 0 & -45 \end{pmatrix} \xrightarrow{C_2 = C_2 - 10C_1} \begin{pmatrix} 3 & 0 \\ 0 & -45 \end{pmatrix} \xrightarrow{R_2 = -R_2} \begin{pmatrix} 3 & 0 \\ 0 & 45 \end{pmatrix}.$$

□

Problem 2. Read the full proof of Smith Normal Form (in Smith Normal Form Supplement under Week 16), and then list any steps which require R to be a PID.

The proof uses that R is a PID in the following places.

First, immediately after defining the function that counts the number of prime factors $\ell(a)$ of an element $a \in R$, this only makes sense in full generality if every element has a factorization into primes. This is true because every PID is a UFD.

Second, the proof uses that every ideal generated by finitely many elements is principal. For example, in Case 2a it says that since R is a PID, $\langle a_1, a_2 \rangle = \langle d \rangle$. This step requires R to be a PID.

Third, the same issue occurs later whenever the proof replaces an ideal generated by several entries with a single generator.

Problem 3. Let

$$R = k[x, y, z].$$

Explain what goes wrong when you try to find a “Smith Normal Form” of

$$\begin{pmatrix} x & y \\ z & x + y \end{pmatrix}.$$

Proof. Let $R = k[x, y, z]$. Notice that R is not a PID. In particular, $\langle x, y, z \rangle$ is not principal since if $\langle x, y, z \rangle = \langle d \rangle$, then d divides x , y , and z . Since x, y, z have no common nonunit factor, d must be a unit. So $\langle d \rangle = R$. But this would imply $\langle x, y, z \rangle = R$, which is impossible since every element of $\langle x, y, z \rangle$ has no constant term.

Now consider $\begin{pmatrix} x & y \\ z & x + y \end{pmatrix}$. The entries generate the ideal $\langle x, y, z, x + y \rangle = \langle x, y, z \rangle$. If a Smith normal form existed by the usual PID algorithm, the first diagonal entry would generate the ideal generated by all entries. But $\langle x, y, z \rangle$ is not principal. Therefore there is no single gcd of the entries obtained as an R -linear combination of $x, y, z, x + y$.

So the first step of the Smith normal form algorithm already fails: we cannot replace the entries by a greatest common divisor using row and column operations. Therefore the usual Smith normal form process does not work over $k[x, y, z]$.

□

Problem 4. Let R be an integral domain with field of fractions $F = Q(R)$, and let M be any R -module.

Prove that the rank of M is the dimension of the vector space $F \otimes_R M$ over F .

Proof. Let $F = Q(R)$. ($\implies$) Suppose $m_1, \dots, m_n \in M$ are R -linearly independent. We prove that $1 \otimes m_1, \dots, 1 \otimes m_n$ are F -linearly independent in $F \otimes_R M$.

Suppose $\sum_{i=1}^n \lambda_i (1 \otimes m_i) = 0$, where $\lambda_i \in F$. Write each $\lambda_i = a_i/s_i$ with $a_i \in R$ and $s_i \in R \setminus \{0\}$. Let $s = s_1 \cdots s_n$. Multiplying by s , we get $\sum_{i=1}^n b_i (1 \otimes m_i) = 0$ for some $b_i \in R$. Equivalently, $1 \otimes \sum_{i=1}^n b_i m_i = 0$. Since the m_i are R -linearly independent, each $b_i = 0$, and therefore each $\lambda_i = 0$. So the pure tensors are F -linearly independent.

($\impliedby$) On the other hand, suppose $1 \otimes m_1, \dots, 1 \otimes m_n$ are F -linearly independent. If $\sum_{i=1}^n r_i m_i = 0$ with $r_i \in R$, then tensoring gives $\sum_{i=1}^n r_i (1 \otimes m_i) = 0$ in $F \otimes_R M$. Since the tensors are F -linearly independent, each $r_i = 0$. This holds since $R_i \cong_R \frac{R_i}{1} \leq_R F$. Therefore, $m_1, \dots, m_n$ are R -linearly independent.

So then R -linearly independent subsets of M correspond to F -linearly independent subsets of $F \otimes_R M$. That is, if $\text{rank } M \leq n$, there exists some R -linearly independent $m_1, \dots, m_n \in M$. If we suppose $\text{rank } F < n$ we then get a contradiction since $1 \otimes m_1, \dots, 1 \otimes m_n$ are F -linearly independent. If M has infinite rank, then we can form an infinite set of R -linearly independent elements in M and a set of F -linearly independent pure tensors corresponding to them. So $\text{rank } M = \dim_F(F \otimes_R M)$. $\square$

Problem 5. Let $R = \mathbb{Z}[x]$ and let $M = \langle 2, x \rangle$.

- (a) Show that $\{2, x\}$ is not a basis for M .
- (b) Show that M has rank 1 but is not free of rank 1.
- (c) Write a presentation for M .

Proof. (a) In $M = \langle 2, x \rangle \subseteq \mathbb{Z}[x]$, we have the nontrivial relation $x \cdot 2 + (-2) \cdot x = 0$. The coefficients x and -2 are both non-zero in $\mathbb{Z}[x]$. So by our equivalence theorem $\langle 2, x \rangle$ cannot be a basis for $\mathbb{Z}[x]$.

(b) Since M is a nonzero submodule of $R = \mathbb{Z}[x]$, and R is an integral domain, any two nonzero elements of M are R -linearly dependent since for any $m, n \in M \setminus \{0\}$ and any we have that $nm - mn = 0$, a non-trivial R -linear combination. So no two elements of M can be linearly independent, and $\text{rank } M \leq 1$ but M does contain a non-zero element 2 which is then linearly independent by itself and so $\text{rank } M = 1$.

Now suppose M were free of rank 1. Then $M = \langle f \rangle$ for some $f \in M$, so M would be a principal ideal. Since $2, x \in M$, we would have $f \mid 2$ and $f \mid x$ in $\mathbb{Z}[x]$. The only common divisors of 2 and x are units, so f would be a unit. Then $M = \langle f \rangle = R$, which is false because $1 \notin \langle 2, x \rangle$. So M is not free of rank 1.

(c) Let $\phi : R^2 \rightarrow M$ be defined by $\phi(a, b) = 2a + xb$. This map is obviously surjective. $2a + xb = 0 \implies xb = -2a$. So $2a = -xb$, and $x \mid 2a$. Since x is prime in $\mathbb{Z}[x]$ and $x \nmid 2$, we get that $x \mid a$ and $a = xh$ for some $h \in R$.

Substituting back into $2a = -xb$ gives $2xh = -xb$. Since $R = \mathbb{Z}[x]$ is an integral domain, we can cancel x to get $2h = -b$. So $b = -2h$.

Therefore $(a, b) = (xh, -2h) = h(x, -2)$, so $\ker \phi \subseteq \langle (x, -2) \rangle$. Then in the other direction we have that $\phi(x, -2) = 2x + x(-2) = 0$. Therefore, $\ker \phi = \langle (x, -2) \rangle$.

Therefore a presentation is $R \xrightarrow{\begin{pmatrix} x \\ -2 \end{pmatrix}} R^2 \rightarrow M \rightarrow 0$. Equivalently, $M \cong R^2 / \langle (x, -2) \rangle$.

□

Problem 6. Let R be a ring and let M be an R -module.

- (a) Prove that if M is projective, then $\text{Tor}(M) = (0)$.
- (b) Prove that a finitely generated module over a PID is projective if and only if it is free.

Proof. (a) Let M be projective. Then there exists an R -module N such that $M \oplus N$ is free. Since $M \oplus N$ embeds into a free module, if $m \in M$ is torsion, then there exists nonzero $r \in R$ such that $rm = 0$. But then $r(m, 0) = (0, 0)$ in the free module $M \oplus N$. Since free modules over a domain are torsion-free, $(m, 0) = 0$, and so $m = 0$. Thus, $\text{Tor}(M) = \langle 0 \rangle$.

(b) If M is free, then M is projective automatically by our equivalence theorem since it is a direct summand of its free self.

On the other hand, suppose M is finitely generated and projective over a PID. Since M is finitely generated over a PID, the structure theorem gives $M \cong R^r \oplus R/(d_1) \oplus \cdots \oplus R/(d_k)$, where each d_i is a nonzero nonunit. By part (a), projective modules are torsion-free, so the torsion summands must vanish. Therefore $M \cong R^r$ and M is free.

□

Problem 7. Let R be a PID which is not a field.

Prove that no finitely generated R -module is injective.

Hint: Recall that $Q \cong Q_1 \oplus Q_2$ is injective if and only if Q_1 and Q_2 are injective.

Proof. We prove that no nonzero finitely generated R -module is injective.

Since R is a PID and not a field, choose a nonzero nonunit $r \in R$. First, R is not injective as an R -module, or we contradict a previous homework problem where we proved that if R is injective over itself then it must be a field.

Next, let $d \in R$ be a nonzero nonunit. We show $R/\langle d \rangle$ is not injective. Consider the inclusion $\langle d \rangle \hookrightarrow R$, and let $h : \langle d \rangle \rightarrow R/\langle d \rangle$ be defined by $h(da) = [a]$. This is well-defined since R is a domain. If $R/\langle d \rangle$ were injective, then h would extend to some $f : R \rightarrow R/\langle d \rangle$. But then $h(d \cdot 1) = [1] = f(d) = df(1) = [0]$ in $R/\langle d \rangle$. But then $1 - 0 = 1 \in \langle d \rangle \implies \langle d \rangle = R$, a contradiction since d is a nonzero nonunit $\implies \langle d \rangle$ is proper. So $R/\langle d \rangle$ is not injective.

Now let M be a nonzero finitely generated R -module. By the structure theorem for finitely generated modules over a PID, $M \cong R^n \oplus R/\langle d_1 \rangle \oplus \cdots \oplus R/\langle d_k \rangle$, where each d_i is a nonzero nonunit.

If M were injective, then every direct summand of M would be injective by a previous homework problem. But each possible nonzero summand is either R or some $R/\langle d_i \rangle$, and we proved neither type is injective.

Therefore no nonzero finitely generated R -module is injective.

□

Problem 8. Let

$$V = \mathbb{R}[x]/(x^4 + 2x^2 + 3x + 1).$$

- (a) Show that V is an $\mathbb{R}$ -vector space with basis $\beta = \{1, x, x^2, x^3\}$.
- (b) Prove that the map $T : V \rightarrow V$ given by multiplication by x is an $\mathbb{R}$ -linear transformation.
- (c) Find the matrix $[T]_{\beta}^{\beta}$.

Proof. (a) Let $V = \mathbb{R}[x]/(x^4 + 2x^2 + 3x + 1)$. Since $x^4 = -2x^2 - 3x - 1$ in V , every element of V can be expressed with a cubic representative, since that relation reduces all powers higher than 3. So then $\{1, x, x^2, x^3\}$ (modulo $x^4 + 2x^2 + 3x + 1$) generates V .

If $a_0 + a_1x + a_2x^2 + a_3x^3 = 0$ in V , then $a_0 + a_1x + a_2x^2 + a_3x^3$ belongs to the ideal $(x^4 + 2x^2 + 3x + 1)$. But a nonzero element of that ideal has degree at least 4, while this polynomial has degree at most 3. So then all $a_i = 0$ otherwise a degree $1 \leq n \leq 3$ polynomial or a constant besides 0 belongs to $(x^4 + 2x^2 + 3x + 1)$, both contradictions. So then by our equivalence theorem, $\beta = \{1, x, x^2, x^3\}$ must be a basis for V over $\mathbb{R}$, and $V \cong \mathbb{R}^4$ as vector spaces.

(b) Let $T : V \rightarrow V$ be defined by $T(v) = xv$. $\forall u, v \in V, \forall c \in \mathbb{R}$, we have $T(u + v) = x(u + v) = xu + xv = T(u) + T(v)$ and $T(cv) = x(cv) = c(xv) = cT(v)$. Therefore T is $\mathbb{R}$ -linear.

(c) We compute the images of the basis elements: $T(1) = x$, $T(x) = x^2$, $T(x^2) = x^3$, and $T(x^3) = x^4$. In V , we have $x^4 = -(2x^2 + 3x + 1) = -1 - 3x - 2x^2$.

$$\text{Therefore, } [T(1)]_{\beta} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, [T(x)]_{\beta} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, [T(x^2)]_{\beta} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \text{ and } [T(x^3)]_{\beta} = \begin{pmatrix} -1 \\ -3 \\ -2 \\ 0 \end{pmatrix}.$$

and therefore

$$[T]_{\beta}^{\beta} = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

□